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Cosmic GPS

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Abstract
The Global Positioning System is something most of us take for granted. Open an app, and you know exactly where you are. But making GPS accurate enough to place you on the right side of a narrow alley requires solving some surprisingly deep problems: the Earth moves, satellites drift, and clocks lie. This article traces how GPS works, starting with the fundamentals of trilateration, working through the relativistic quirks that complicate timing, and finally landing on the unexpected hero of the story... quasars. They are supermassive black-hole-powered galactic nuclei, billions of light-years away, serve as perfectly stationary anchors that allow astronomers to detect even the subtlest shifts in Earth's orientation and keep GPS honest.

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Introduction

Imagine being stranded somewhere without access to Google Maps to help pinpoint your location. Scary, right? Well, this is just one of the many reasons the Global Positioning System (GPS) is vital to us. But how did we come up with such an idea? How does it work? More importantly, how does it work so precisely? After all, one can spot oneself in even the narrowest alley.

The idea was realized through a connecting-the-dots process, which is why we will first discuss different, quite unrelated "dots" and then connect them to see how they give us this powerful navigation tool.

1. "Dots": The Motivation

Dot-1: The story begins with this technique known as Very-Long-Baseline Interferometry (VLBI), which captures radio images in the sky. The important thing to note here is that these images are not captured by just one telescope, but by many. Combining their data to achieve higher resolution is what the term "Interferometry" in the name VLBI refers to.

Picture this: Two telescopes separated by some distance won't receive the data "simultaneously". There will always be a slight delay. Say the telescopes captured the image of an object. By comparing the data and using the difference in acquisition times, the distance between the two telescopes and the angle of inclination of the Earth relative to that object can be estimated. This means that if we had two telescopes across the globe imaging the same object over a short time interval, we could estimate the distance between them and, hence, the size of the Earth. And this is what was done as well! Imaging via VLBI began as early as the 1950s, and various radio sources, including quasars, were captured. [1]

Dot-2: A quasar (quasi-stellar radio source) is an extremely luminous and distant, active galactic nucleus, powered by a supermassive black hole accreting matter. Quasars emit energy across the electromagnetic spectrum and often outshine their host galaxies [2]. These quasars are so far away that effectively they are stationary with respect to us, i.e., if there is a quasar at a declination of 0.5° below Sirius today, it would remain there even after 5 years.

Now, at one point in time, using our Radio Telescopes, we knew the positions of a lot of these Quasars. We can use the concept of Celestial Sphere (Fig 1), which is a projection of all the objects in the observable universe on the surface of a sphere concentric with the Earth, to establish an instantaneous, one-to-one mapping between these astrophysical structures and a specific location on Earth. This is the motivation behind using VLBI for GPS. But knowing where you are relative to the cosmos solves only half the problem; you also need a way to calculate your exact position on the ground. That's where our third dot comes in.

Dot-3: The final dot is Trilateration [3]. Think of it this way: if someone tells you they are 5 km from a coffee shop, you know they're somewhere on a circle of radius 5 km centered on the coffee shop. Add a second landmark, say, 3 km from a park, and now you can narrow down their location to one of two intersection points between two circles. A third landmark pins you to a single point. When we apply the same idea in 3D (Earth system), each distance measurement defines a sphere instead of a circle, and three spheres are enough to narrow you down to (at most) two points, as illustrated in Fig 2. One of these points is usually underground and can be discarded. The requirement for four satellites, however, still persists due to the bias term, which we'll get to in a moment.

2. The Connection

So, there are three components involved. The satellites, the Ground Stations, and the Receiver, like your phone. Imagine the satellites orbiting Earth like a spherical shell. The satellites constantly broadcast signals, like lighthouses. The receiver catches these signals and solves the following equation:

$$(x-x_s)^2+(y-y_s)^2+(z-z_s)^2=c^2(\Delta t-t_b)^2$$

where x,y,z are the receiver coordinates, x_s,y_s,z_s are the satellite coordinates (which, by our assumption, we already know), Δt is the time delay in the signal, c is the speed of light and t_b is the 'bias'. The above equation is taken from the theory of special relativity and relates the distance an (electromagnetic) signal travels (left-hand side) to the time elapsed between its emission and reception (right-hand side). The presence of 'c', the speed of light, is the factor that converts the time quantity into a spatial quantity. The importance of the bias term can be understood by the following explanation. Imagine your wristwatch is running slightly slow; every time you check it, your sense of "now" is off by a tiny, incremental amount. Something similar happens here. The receiver's clock is far less precise than the atomic clocks on satellites, and even a microsecond of error translates to ~ 300 meters of positional error. On top of that, gravity is slightly weaker at the satellite's altitude, which causes its clock to tick marginally faster than one on the ground (yes, General Relativity is very much at play here!). The bias term t_b accounts for the accumulated clock mismatch. Since t_b is an extra unknown, we need a fourth equation, hence, four satellites instead of three [4].

But it turns out that our assumption isn't good enough for such precise calculations. Practically, even the tiniest movements of Earth with respect to the satellites, like precession (which, mind you, is a 26,000-year cycle) or shifting of the tectonic plates or the movements of satellites with respect to the Earth, like those due to solar radiation pressure, can cause this positioning system to fall apart. It won't provide the accurate locations it was meant to. This is exactly where quasars come into play.

Notice that x_s,y_s,z_s are in the Earth's own reference frame. To resolve the above issue, we need to shift to an inertial frame. The VLBI helps us construct this frame, called 'The International Celestial Reference Frame (ICRF)', which helps us in the following way. Theoretically, the Celestial Sphere, consisting of only the 212 quasars, should only rotate about the Earth's spin axis at a constant rate. If the axis changes even slightly, the rotation rate changes, or some observatories see the quasars at a different angle than they were a few days ago, and we know the Earth moved with respect to the satellites. The data for this motion (which goes by the name Earth Orientation Parameters, EOP) is calculated using VLBI.

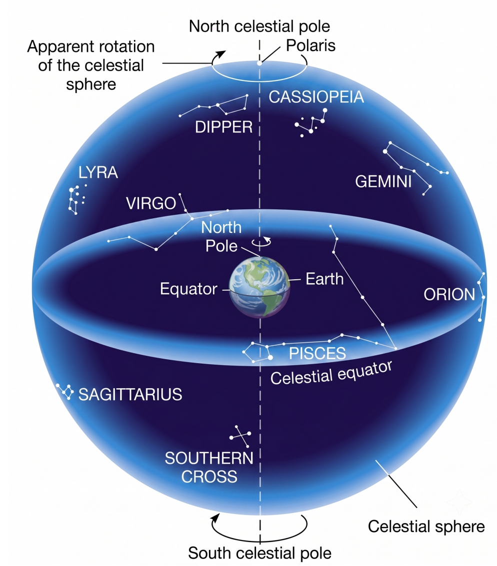

Figure 1. The Celestial Sphere. Image credits: [6]

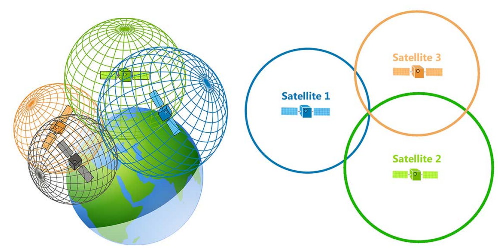

Figure 2. Visualization of Trilateration in 3D (left) and 2D (right). Image credits: [7]

The new satellite coordinates are estimated in the Earth's frame and sent to the satellites, which then broadcast them. If the satellites move, we reverse the process to obtain the precise satellite coordinates again, which the U.S. Space Force handles.

If you've noticed, this entire system is built on the assumption that quasars themselves must not move, or there would be no ICRF. They make good anchor points, which is precisely why we chose extremely radio-bright objects like these rather than just any star. Even though they are invisible to the human eye, they are such strong radio sources that astronomers can (easily?) identify and track them with radio telescopes [5]. See the Quasar Projection Map captured at some particular instant of time in Fig 3.

The marriage of VLBI and GPS didn't happen overnight. In the early 1980s, as GPS satellites were being launched, NASA's Jet Propulsion Laboratory began building a dedicated ground tracking network, deliberately placing the first stations alongside existing VLBI sites. The logic was simple: those sites already had precise location data and the communications infrastructure needed to anchor the GPS reference frame to the celestial one. That chain has grown steadily ever since, and

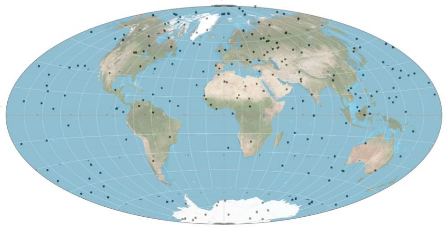

Figure 3. Quasar projection map fusion by Daniel Cummings, quasar locations courtesy NASA.

today it spans more than 80 receivers worldwide, making it the largest centrally managed geodetic GPS tracking network on the planet [8]. In short, every time your phone drops a pin on a map, it is quietly relying on a chain that runs all the way back to radio telescopes staring at billion-light-year-distant black holes.

3. Summary

There you have it, the unlikely chain that makes GPS work. A technique invented to image distant radio sources is used to measure the Earth's own wobbles. Objects so far away that they appear frozen in the sky become the most reliable fixed points we have. A system designed for military navigation ends up guiding you to the nearest coffee shop. The next time your phone tells you to turn left in 200 meters, take a moment to appreciate that the instruction traces back to a supermassive black hole, billions of light-years away, quietly doing its small part to keep you from getting lost!

References

[1] Chris Smith. NASA Scientific Visualization Studio — Using Quasars to Measure the Earth: A Brief History of VLBI — svs.gsfc.nasa.gov. https://svs.gsfc.nasa.gov/10964/.

[2] Quasar - geodesy.science - IAG website — geodesy.science. https://geodesy.science/glossary/quasar/.

[3] Trilateration - Wikipedia — en.wikipedia.org. https://en.wikipedia.org/wiki/{T}rilateration.

[4] How Does GPS Work? — NASA Space Place – NASA Science for Kids — spaceplace.nasa.gov. https://spaceplace.nasa.gov/gps/en/.

[5] GPS uses Quasars - Star In A Star — starinastar.com. https://starinastar.com/gps-uses-quasars/.

[6] Pre-lab #4 The Celestial Globe — physics.mtsu.edu. https://physics.mtsu.edu/prelab3_celestial_globe/.

[7] GISGeography. How GPS Receivers Work - Trilateration vs Triangulation - GIS Geography — gisgeography.com. https://gisgeography.com/trilateration-triangulation-gps/.

[8] NASA Brings Accuracy to World’s Global Positioning Systems — NASA Spinoff — spinoff.nasa.gov. https://spinoff.nasa.gov/{S}pinoff2019/ps_1.html.